ENGN 2350: Data-Driven Design & Analysis of Structures & Materials

SAMPLE OF MIDTERM EXAM – SOLUTION

Brown University, School of Engineering

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Exam to be solved during 50 minutes

There are twenty (20) multiple choice questions. For each, **select only one answer**.

Each question is worth **5 points** and no partial credit will be attributed.

This exam is closed book, except for the formula sheet of the course. You do not need a calculator and you cannot use a computer.

Name (**PRINT**): _____

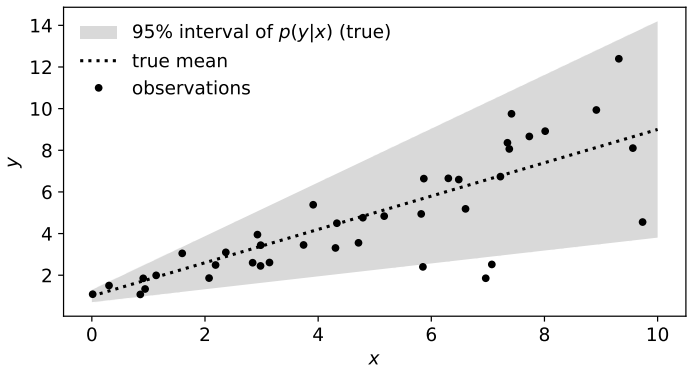
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Multiple choice questions

1. The figure shows observations, the true mean of the process that generated them and the 95% interval of its true observation distribution. No model is shown. Which statement is correct?



   A. We can observe homoscedastic aleatoric uncertainty and epistemic uncertainty.

   B. We can observe homoscedastic aleatoric uncertainty, but not epistemic uncertainty.

   C. We can observe heteroscedastic aleatoric uncertainty and epistemic uncertainty.

   D. We can observe heteroscedastic aleatoric uncertainty, but not epistemic uncertainty.

   The band widens with x (heteroscedastic noise); no model was fitted, so there is no uncertainty about a model's parameters (epistemic).

2. Let $z \sim \mathcal{N}(0,1)$ and $y = z^2$. Which statement is correct?

   A. $\mathrm{Cov}[y,z] = 0$, so y and z are independent.

   B. $\mathrm{Cov}[y,z] = 1$, because y grows with $|z|$.

   C. $\mathrm{Cov}[y,z] = 0$, but y and z are not independent.

   D. $\mathrm{Cov}[y,z]$ is undefined, because y is not Gaussian.

   Sheet: $\mathrm{Cov}[y,z] = \mathbb{E}[yz] - \mathbb{E}[y]\,\mathbb{E}[z]$, here $= \mathbb{E}[z^3] - \mathbb{E}[z^2]\,\mathbb{E}[z]$.
   $\mathbb{E}[z] = 0$, and $\mathbb{E}[z^3] = \int z^3\,\mathcal{N}(z|0,1)\,dz = 0$ (sheet: $\mathbb{E}[g(z)] = \int g\,p$): z^3 is odd and the pdf is symmetric about 0, so the positive and negative parts cancel.
   So $\mathrm{Cov}[y,z] = 0$, yet y is a function of z (knowing z fixes y): uncorrelated does not mean independent. The sheet states only one direction: independent $\Rightarrow$ uncorrelated.

3. Let $z \sim \mathcal{U}(0,1)$ and $y = z^2$. What is the pdf $p(y)$ on $(0,1)$?

   A. 1

   B. $\sqrt{y}$

   C. $2y$

   D. $1/(2\sqrt{y})$

   Change of variables (sheet) with $z = g(y) = \sqrt{y}$: $p(y) = p_z(\sqrt{y})\,|dg/dy| = 1 \cdot \frac{1}{2\sqrt{y}}$. $2y$ uses the derivative of the forward map $y = z^2$ instead of the inverse.

4. With $p(y|z) = \mathcal{N}(y|z^2, 1)$ and the prior $p(z) = \mathcal{N}(z|0, 1)$, which statement about the posterior $p(z|y)$ is correct?

   A. It is Gaussian, because the product of two Gaussians is proportional to a Gaussian.

   B. It is not Gaussian: as a function of z, the likelihood is not a Gaussian.

   C. It is Gaussian, but its variance has no closed form.

   D. It becomes Gaussian only after observing many values of y.

   Bayes: $p(z|y) \propto p(y|z)\, p(z)$.

   As a function of z, the likelihood is $p(y|z) \propto \exp[-(y - z^2)^2/2]$. Its exponent is a polynomial of degree 4 in z, not 2, so it is **not** a Gaussian in z.

   The product rule (sheet: product of two Gaussians in $\mathbf{x}$) needs both factors to be Gaussians in the same variable z. That happens when the mean of the observation distribution is linear in z (e.g. $\mu_{y|z} = wz + b$): then $(y - wz - b)^2$ is quadratic in z.

   Here the posterior $\propto \exp[-(y - z^2)^2/2 - z^2/2]$ is not Gaussian; for large y it even has two peaks, near $z = \pm\sqrt{y}$.

5. Measurements are $y = z + \epsilon$ with $\epsilon \sim \mathcal{N}(0, 3)$ known, and the prior on z is Uniform. $N = 3$ measurements have mean 10. Which interval contains a new measurement with 95% probability?

   A. $10 \pm 1.96 \cdot 1$

   B. $10 \pm 1.96 \cdot \sqrt{3}$

   C. $10 \pm 1.96 \cdot 2$

   D. $10 \pm 1.96 \cdot 3$

   Write it as a linear model (sheet: linear models). y does not depend on any input x, and there is a single weight, $w = z$, with basis $\phi = 1$.

   For the $N = 3$ measurements: $\mathbf{\Phi} = [1, 1, 1]^T$ (a 3×1 matrix), $\mathbf{\Lambda} = 3\,\mathbf{I}_3$ (the noise is known), and the Uniform prior means $\overset{<}{\mathbf{\Sigma}}{}_w^{-1} \to 0$.

   Posterior (sheet): $\overset{>}{\Sigma}_w = \left(\overset{<}{\Sigma}{}_w^{-1} + \mathbf{\Phi}^T \mathbf{\Lambda}^{-1} \mathbf{\Phi}\right)^{-1} = \left(0 + \frac{1}{3}(1+1+1)\right)^{-1} = 1$, and $\overset{>}{\mu}_w = \overset{>}{\Sigma}_w\, \mathbf{\Phi}^T \mathbf{\Lambda}^{-1} \mathbf{y} = \frac{1}{3}(y_1 + y_2 + y_3) = \bar{y} = 10$.

   PPD of a new measurement (sheet): $(\sigma^*)^2 = \sigma^2 + \phi\, \overset{>}{\Sigma}_w\, \phi = 3 + 1 = 4$, so $\sigma^* = 2$ and the 95% interval is $10 \pm 1.96 \cdot 2$. $10 \pm 1.96 \cdot 1$ is the credible interval for z (it leaves out the noise of the new measurement); $10 \pm 1.96\sqrt{3}$ is the plug-in PPD (it leaves out the uncertainty about z).

6. A quantity z has the prior $p(z) = \mathcal{N}(z|0, 4)$. One measurement $y = 2$ is taken, with $p(y|z) = \mathcal{N}(y|z, 4)$. What is the posterior $p(z|y)$?

   A. $\mathcal{N}(z|1, 2)$

   B. $\mathcal{N}(z|2, 4)$

   C. $\mathcal{N}(z|1, 4)$

   D. $\mathcal{N}(z|2, 2)$

   Bayes: $p(z|y) = p(y|z)\, p(z)/p(y)$, with $p(y = 2|z) = \mathcal{N}(2|z, 4) = \mathcal{N}(z|2, 4)$ as a function of z, and $p(z) = \mathcal{N}(z|0, 4)$.

   Product of two Gaussians in z (sheet): $\mathcal{N}(z|2, 4)\, \mathcal{N}(z|0, 4) = \mathcal{N}(2|0,\ 4+4)\, \mathcal{N}(z|c, C)$. The first factor does not depend on z: it cancels with the marginal likelihood $p(y)$ (the posterior must integrate to 1), so it never needs to be computed.

   $C = (1/4 + 1/4)^{-1} = 2$, then $c = C\,(2/4 + 0/4) = 2 \cdot \frac{1}{2} = 1$.

   Posterior: $\mathcal{N}(z|1, 2)$.

7. For a prior that is nonzero everywhere, what is the influence of the prior on the posterior predictive distribution in the limit of infinitely many data points?
   A. It tends to alleviate overfitting.
   B. It has no influence.
   C. It tends to alleviate underfitting.
   D. It cannot be determined without more information.

   With infinitely many data the likelihood dominates the posterior wherever the prior is nonzero: the prior's effect fades away.

8. You derived the posterior of a model, but the integral of the posterior predictive distribution has no analytical solution. What is a reasonable way to proceed?
   A. Find the location of the minimum of the negative log posterior and recompute the posterior with Bayes' rule.
   B. Find the location of the maximum of the posterior (its mode) and plug it into the observation distribution.
   C. Find the location of the maximum of the prior (its mode) and plug it into the observation distribution.
   D. Use the prior instead of the posterior in the integral.

   Replace the posterior by a Dirac delta at its maximum (MAP): the PPD becomes the observation distribution at that point (plug-in).

9. Which observation distribution does **not** lead to a linear regression model?
   A. $\mathcal{N}(y|\mu_{y|z} = w_0^2 + w_1 x,\ \sigma^2)$
   B. $\mathcal{N}(y|\mu_{y|z} = w_0 + w_1 \cos x,\ \sigma^2)$
   C. $\mathcal{N}(y|\mu_{y|z} = w_0 x + w_1 x^3,\ \sigma^2)$
   D. $\mathcal{N}(y|\mu_{y|z} = w_1 e^x + w_2 \tanh x^2,\ \sigma^2)$

   Linear regression must be linear in the parameters: w_0^2 is not. The other three are linear in the weights ($\cos x$, x^3, e^x, $\tanh x^2$ are fixed basis functions).

10. Least Squares with $M = 3$ weights is fitted to $N = 12$ points and gives $\hat{\sigma}^2_{\text{mle}} = 1.5$. What is the unbiased estimate of the noise variance?
    A. 1.125
    B. 1.5
    C. 18/11
    D. 2

    $\hat{\sigma}^2_{\text{mle}} = \text{RSS}/N$, so RSS $= 18$; with $\mathbb{E}[\text{RSS}] = (N - M)\,\sigma^2$ (sheet: Least Squares) the unbiased estimate is $18/9 = 2$. The MLE is biased low because the fit absorbs part of the noise.

11. You choose among 3 values of the Ridge strength α and 4 polynomial degrees by 5-fold cross-validation on the training set. How many models are fitted before the final refit?

    A. 12

    B. 17

    C. 20

    D. 60

    $3 \times 4 = 12$ combinations, each fitted 5 times: 60.

12. Measurements have a known noise standard deviation of 0.3. An agent reports a mean squared error of 0.01 on 200 noisy test measurements. What do you conclude?

    A. The model is excellent: it predicts better than the noise level.

    B. Nothing unusual: 0.01 is below the noise standard deviation 0.3, as expected.

    C. Something is wrong, e.g. a leak of test points: no model averages below 0.09.

    D. The noise level is wrong: the test MSE estimates the noise standard deviation.

    Noise + bias^2 + variance (sheet): on noisy measurements the expected test MSE is at least $\sigma^2 = 0.3^2 = 0.09$; with 200 points, 0.01 is implausible.

13. In Ridge regression with a fixed noise level, the prior standard deviation of the weights is doubled. What happens?

    A. α doubles, and the weights are pulled more strongly toward zero.

    B. α is unchanged, because it is set by the data alone.

    C. α becomes 4 times larger, and the weights move toward zero.

    D. α becomes 4 times smaller, and the weights move toward Least Squares.

    Sheet (Ridge): $\alpha = \sigma^2 / \overset{<}{\sigma}{}^2_w$, so doubling $\overset{<}{\sigma}_w$ divides α by 4: a weaker prior, and the estimate moves toward Least Squares ($\alpha \to 0$).

14. Why is the Ridge point estimate of the weights easier to find than the type-II maximum likelihood estimate of the hyperparameters of a Gaussian process?

    A. A Gaussian process has many more parameters to estimate.

    B. Ridge is quadratic in $\mathbf{w}$; the marginal likelihood is not convex in $\boldsymbol{\theta}$.

    C. The marginal likelihood can only be evaluated on test data.

    D. Ridge has no hyperparameters to optimize.

    Ridge: closed form $(\boldsymbol{\Phi}^T\boldsymbol{\Phi} + \alpha\mathbf{I}_0)^{-1}\boldsymbol{\Phi}^T\mathbf{y}$ (sheet). Log marginal likelihood (sheet): $\boldsymbol{\theta}$ inside $\mathbf{K}_y^{-1}$ and $\log|\mathbf{K}_y|$, no closed form, local optima.

15. In the noise step of the staggered scheme, each squared residual is increased by the epistemic variance of the mean at that point. Why?

A. The epistemic variance is the measurement noise, which the residual does not contain.

B. It makes the noise step convex, so that Newton's method converges.

C. Without it, the noise is underestimated: the fitted mean is itself uncertain.

D. It removes the bias of the mean, so that the PPD mean becomes unbiased.

Sheet (staggered scheme): $\tilde{r}_n^2$ adds $\mathbb{V}[\mu_{y|z}(\mathbf{x}_n)]$; the plain residual of an uncertain mean fitted to the same points is too small.

16. A Gaussian process with an RBF kernel is fitted to noiseless data at $x = 0$ and $x = 1$. If the length scale l is halved, with everything else fixed, the PPD standard deviation at $x = 0.5$:

A. increases, because $x = 0.5$ becomes less correlated with the data.

B. decreases, because a shorter length scale makes the fit more flexible.

C. stays the same, because it depends only on the amplitude s.

D. becomes zero, because the data are noiseless.

Sheet (GP PPD): the variance is $k(\mathbf{x}^*, \mathbf{x}^*) - \mathbf{k}^{*T}\mathbf{K}^{-1}\mathbf{k}^*$; with a shorter l the entries of $\mathbf{k}^*$ (RBF) shrink, less is subtracted, and the band between the points widens. It is zero only at the training inputs.

17. A Gaussian process with zero prior mean and an RBF kernel with $s = 1$ is fitted to one noiseless observation, $y = 2$ at $x = 0$. At a new input x^*, $k(x^*, 0) = 0.5$. What is the PPD at x^*?

A. $\mathcal{N}(y^*|1,\ 0.5)$

B. $\mathcal{N}(y^*|1,\ 0.75)$

C. $\mathcal{N}(y^*|2,\ 0.75)$

D. $\mathcal{N}(y^*|1,\ 1)$

Sheet (GP PPD) with $N = 1$, $\mathbf{K} = k(0,0) = s^2 = 1$: $\mu^* = 0.5 \cdot 1^{-1} \cdot 2 = 1$ and $(\sigma^*)^2 = 1 - 0.5 \cdot 1^{-1} \cdot 0.5 = 0.75$. Variance 1 is the prior; mean 2 copies the data point.

18. A Gaussian process with zero prior mean uses the kernel $4^2 \exp\left(-\|\mathbf{x} - \mathbf{x}'\|^2/(2l^2)\right)$ plus a White kernel with $s_2 = 3$. Far from all training inputs, what is the standard deviation of its posterior predictive distribution?

A. 3

B. 4

C. 5

D. 7

The kernel is 4^2 RBF $+ 3^2\,\delta_{ij}$ (sheet: kernels, White).

Far from all training inputs $\mathbf{k}^* \to \mathbf{0}$, so the PPD (sheet) reduces to the prior: $(\sigma^*)^2 = k(\mathbf{x}^*, \mathbf{x}^*) = 4^2 + 3^2 = 25$.

Variances add, standard deviations do not: $\sigma^* = \sqrt{25} = 5$, not $4 + 3 = 7$.

19. With 5 training points, two very different (length scale, noise level) pairs give almost the same marginal likelihood. The best interpretation:

A. The data are too few to determine the hyperparameters: both explanations are plausible.

B. The optimizer failed: with more restarts one pair will clearly win.

C. The pair with the smaller noise must be preferred, as it fits the data better.

D. The kernel is invalid: a valid kernel has a single maximum.

Data fit and complexity (sheet: log marginal likelihood) balance almost equally for both pairs: with 5 points the data cannot decide.

20. A Gaussian process uses an RBF kernel (s, l) plus a White kernel (s_2). Which statement is correct?

A. s, l are weights fitted by Least Squares; s_2 is a hyperparameter.

B. The model has no parameters, so it needs no model selection.

C. s, l, s_2 are hyperparameters; the weights are integrated out.

D. s, l, s_2 are fixed by the kernel and cannot be learned.

Hyperparameters set the prior over functions (sheet: kernels, type-II estimates); the infinitely many weights are integrated out.